1. Simulation process. Initialize with a random distribution of a pair of average concentration and internal energy $(\bar{c}, \bar{E})$.

2. A coupled partial melting Stokes-Darcy system. Strong form

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0\phi_f^{1+\Theta}}{\mu_f}\nabla(\phi_f^{-1/2}\tilde{q}_f) &= 0\\
\mu_s\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\tilde{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0\\
\nabla q - \nabla\cdot\hat{\sigma}(\mathbf{v}_s) &= -(1-\phi_f)\rho_r\mathbf{g}\\
\mu_s\nabla\cdot\mathbf{v}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\tilde{q}_f - \phi_f^{1/2}q) &= 0
\end{aligned}
\tag{2.1}
$$

List SI units
q $=$ pascal.
$\mathbf{v}$ $=$ m/s
Non dimensionalization choice

- $\check{\mathbf{v}} = \frac{\tilde{\mathbf{v}}_r}{k_0\rho_r g_y/\mu_f}$
- $\check{q} = \frac{q}{\rho_r g_y l_0}$
- $\check{x} = \frac{x}{l_0}$
- $l_0 = (\frac{k_0\mu_s}{\mu_f})^{1/2}$

The resulting non dimensionalized form is therefore

$$
\begin{aligned}
\check{\mathbf{v}}_r + \phi_f^{1+\Theta}\nabla(\phi_f^{-1/2}\check{q}_f) &= 0\\
\phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0\\
\nabla\check{q} - \nabla\cdot\hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\rho_r\hat{\mathbf{g}}\\
\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0
\end{aligned}
\tag{2.2}
$$

scaled nondimensionalized weak form let $\breve{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla\cdot\mathbf{v}\mathrm{I}$

$$
\begin{aligned}
(2\phi_s\breve{\sigma}(\mathbf{v}_{s,h}), \nabla\Psi_s) - (\check{q}_h, \nabla\cdot\Psi_s) &= -(\phi_s\check{\mathbf{f}}, \Psi_s)\\
(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h}, \omega\right) &= 0\\
(\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) &= 0\\
(\hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s}\check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\right) &= 0
\end{aligned}
\tag{2.3}
$$

2.1. Variational form and discretization.

2.1.1. Stokes and variational formulation.

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{2.4}$$

weak form

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega = (\mathbf{f}, \mathbf{v})_\Omega + &< \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau >_{\Gamma_i} \\ &+ < \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu >_{\Gamma_i} - < \nabla \mathbf{u}_0, \nabla \mathbf{v} >_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) = 0 \end{aligned} \tag{2.5}$$

let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have traction boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. Traditionally a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$ is employed here.

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} = (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + < \mathbf{s}, \mathbf{v}_h >_{\Gamma_{i,h}} - < \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h >_{\Gamma_{u,h}} \\ (\nabla \cdot \mathbf{u}, \omega) = 0 \end{aligned} \tag{2.6}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

2.2. Darcy.

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{2.7}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega = (\mathbf{g}, \mathbf{v})_\Omega - < p_0, v_n >_{\Gamma_i} - < \mathbf{u}_0, \mathbf{v} >_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) = 0 \end{aligned} \tag{2.8}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

2.2.1. A degenerate Darcy test problem.

$$\begin{aligned} \mathbf{v} + \phi \nabla (\hat{\phi}^{-1/2} p) &= f \\ \hat{\phi}^{-1/2} \nabla \cdot (\phi \mathbf{v}) &= g \end{aligned} \tag{2.9}$$

We manufacture a solution with degenerate porosity in the target physical domain $x, y \in [-1, 1]$.

$$\phi = \begin{cases} \cos^m(n(x^2+y^2)) & x^2 + y^2 < \pi/2n \\ 0 & \text{otherwise} \end{cases} \tag{2.10}$$

where m is an even number, and $\pi/2n < 1$ another choice of test case is shown as follows:

$$\phi = \begin{cases} 0 & x \leq 0 \\ \phi_+ x^4 & x > 0 \end{cases} \tag{2.11}$$

Define an arbitrary velocity and pressure

$$\begin{aligned} \mathbf{v} &= [\frac{x}{5}, -y]^T \\ p &= \hat{\phi}^{1/2} \end{aligned} \tag{2.12}$$

The source term is calculated as follows:

$$f = \mathbf{v} \tag{2.13}$$

and the divergence term is:

$$\begin{aligned} g &= \phi^{-1/2}\phi_+ \nabla \cdot [\frac{x^5}{5}, -x^4 y]^T \\ &= \phi^{-1/2}\phi_+(x^4 - x^4) = 0 \end{aligned} \tag{2.14}$$

a divergence free test case

2.2.2. Modified BR element. The Bernardi-Raugel element has been researched extensively in previous literature [2]. We state that our new function space satisfies the following hypotheses:

- For any q in $H^m(\Omega)$

3. Linear system and Uzawa type algorithm. Boundary conditions are already included in the linear system formulation. Original linear system:

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix} \tag{3.1}$$

This is a double saddle point system. Symmetrically permutate original linear system and form a enlarged saddle point system.

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix} \tag{3.2}$$

(3.3)

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} -\mathbf{f}_d \\ -\mathbf{g}_d \end{bmatrix}$$

In this newly formed system, the $\mathbf{A}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [3], [5].

Consider the linear system of equation:

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix} \tag{3.4}$$

Algorithm 3.1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{Q}^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$

In the previous literature [5], $\mathbf{Q}$ is approximated with $\tau\mathbf{I}$, which works well with the stokes part. However, it converges very slow with the same approximation when applied to the Darcy part. We need a better preconditioner approximating inverse of the corresponding schur complement. We use MINRES algorithm [1] due to schur complement being symmtetric and indefinite.

Algorithm 3.2 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

Based on the above algorithm, we proposed a new block preconditioner for our coupled linear system.

$$\mathbf{Q}^{-1} = \begin{bmatrix} \mathbf{S}_s^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_d^{-1} \end{bmatrix} \tag{3.5}$$

where the schur complement matrices are approximated with MINRES method.

3.1. Convergence and stability analysis. Uzawa iteration is in fact a fixed point iteration procedure. Usually, uzawa iteration convergences slowly.

3.1.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [7]

4. Test cases. We test the linear system a little bit different from the linear system above.

4.1. Constant porosity and balanced pressure (divergence free velocity field).

4.2. Const porosity and unbalanced pressure.

$$\check{q} = 0 \qquad \check{q}_f = 2(x+y) \qquad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \qquad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \tag{4.1}$$

$$\nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y) \tag{4.2}$$

$$\mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \qquad (\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I} = \frac{2\phi_f^{1/2}}{1-\phi_f}\begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix} \tag{4.3}$$

$$\begin{aligned} \tilde{\mathbf{v}}_r + \phi_f^{1/2}\nabla \check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f}\begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2}\begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2}\nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}\check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f}2(x+y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}\check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\ -2(1-\phi_f)\nabla \cdot \sigma(\check{\mathbf{v}}_s) = -2(1-\phi_f)\nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s &- \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathrm{I}) = -\frac{8\phi_f^{1/2}}{3}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned} \tag{4.4}$$

5. Hyperbolic Equation System. In the simulatin of partial melting system, there arises multidimensional hyperbolic equations systems.

$$\mathbf{U}_t + \nabla \cdot [F(\mathbf{U}) - A(\mathbf{U}\nabla\mathbf{U})] = 0$$

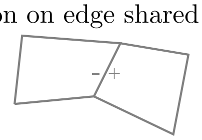
$\mathbf{U}$ consists of two variables, average concentration of one species $\bar{c}_1$ and average total internal energy $\bar{E}$.

$$\bar{c}_t + \nabla \cdot (\mathbf{v}_e\bar{c} - \kappa D\nabla\bar{c}) = 0$$

The finite volume discretization can be written as:

$$empty$$

where $\kappa = \frac{\phi_f c_f}{\bar{c}}$ need definition on edge shared between two cells.



There are two κ values defined on both sides of the shared edge.

$$\kappa^{\pm} = \frac{\phi_f c_f^{\pm}}{\bar{c}^{\pm}}$$

where ϕ_f has definition on the edge.

6. Eutectic phase behavior. Mid-ocean ridge forms at the place where temperature is generally lower. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution.

On a typical eutectic phase behavior diagram, we can locate any phase situation using temperature and composition information. We follows phase behaviors details described in [6]. We made some reasonable adjustments for our mantle system.

There are few key elements that allow practical calculation of phase region and corresponding volumetric fractions.

- Fluid formed by eutectic melting has a constant composition of opx denoted as X_e. The eutectic composition can be related to pressure and denoted as $X_e(P)$.
- Melting temperature relates to pressure. Decompression melting is accounted for in our model by relating eutectic melting temperature with pressure change.
- Eutectic melting is of the most insterest. The temperature where eutectic melting happens has much lower temperature then actual melting temperature.

We define following dimensionless bulk variables following [6]. We denote volumetric fraction of three phases as olivine ϕ_1, opx ϕ_2 and melt ϕ_f. We are tracking opx in the mixture for simulation. Stefan number is defined as the ratio of the sensible heat to latent heat, Ste $= \frac{c_p \Delta T}{L}$. We define dimensionless temperature with eutectic temperature as $\mathcal{T} = \frac{T - T_e}{T_{\text{MAX}} - T_e}$, and dimensionless pressure as $\mathcal{P} = \frac{P - P_0}{P_0}$, where P_0 represents standard atmospheric pressure. We use melting temperature of olivine at standard atmospheric pressure as 2053 K, and 1753 K for orthopyroxene (opx). Olivine has density range of 3.2 to 4.3 g/cm^3 and orthopyroxene has similiar density ranging from 3.2 to 4 g/cm^3. We define a dimensionless composition $\chi_i = \frac{X}{X_e}$ where $i \in \{1, 2, f\}$ and bulk composition defined as $\chi = \sum \phi_i \chi_i$. The eutectic melting composition is now normalized as $\chi_e = X_e/X_e = 1$.

$$
\begin{aligned}
C &= \phi_2 \rho_2 + \phi_f \rho_f \chi_f(\mathcal{T}, \mathcal{P}) \\
H &= \phi_1 \rho_1 c_{p,1} \mathcal{T} + \phi_2 \rho_2 c_{p,2} \mathcal{T} + \phi_f \rho_f \left(L_f + c_{p,f} \mathcal{T} \right)
\end{aligned} \tag{6.1}
$$

Where the melting latent heat can be affected by multiple things. We can approximate the latent heat of the mixture as

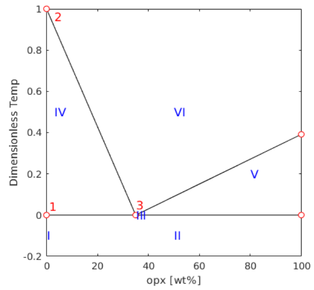
$$
L_f = L_1(1 - \chi_f(\mathcal{T}, \mathcal{P})) + L_2 \chi_f(\mathcal{T}, \mathcal{P})
$$

. We can nondimensionalize bulk composition and dimensionless bulk enthalpy as:

$$
\begin{aligned}
\mathcal{C} &= C/\rho_1 \\
\mathcal{H} &= H/(c_{p,1} \rho_1)
\end{aligned} \tag{6.2}
$$

We denote nondimensionalized density and specific heat as $\rho_{i,j}$ and $c_{p,i,j}$.

The eutectic phase diagram can be simplified as:



6.1. Primary phase split. In mantle, we mostly have olivine and orthopyroxene (opx) mixture in the rock. Like eutectic melting system e.g. salt and ice, the presence of one phase would largely lower the melting temperature of another phase to the eutectic melting point. We now decompose different phase regions in details We are tracking bulk concentration of orthopyroxene (opx) in our calculation. We identify following 6 regions with different phase assemblages:

I Single-phase sub-solidus : olivine
II Two-phase sub-solidus: olivine + opx
III Three-phase eutectic: olivine + opx + melt
IV Two-phase super-eutectic : olivine + melt
V Two-phase super-eutectic : opx + melt
VI Single-phase super-liquidus : melt

6.1.1. Region I. Pure olivine. The conditions to be in region I are

$$\mathcal{C} = 0 \quad \mathcal{H} \leq 1 \tag{6.3}$$

Hence, the volumetric fraction and temperature can be calculated simply as:

$$\phi_1 = 1, \quad \phi_2 = \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \mathcal{H} \tag{6.4}$$

6.1.2. Region II. Two-phase sub-solidus: olivine + opx. In this phase region, temperature hasn't reached eutectic melting temperature.

$$\phi_2 = \frac{\mathcal{C}}{\rho_{2,1}} \quad \phi_1 = 1 - \phi_2 \quad \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \frac{\mathcal{H}}{\phi_1 + \phi_2 \rho_{2,1} c_{p,2,1}} < 0 \tag{6.5}$$

6.1.3. Region III. In this three-phase eutectic region, all three phases are present. The temperature is fixed at the eutectic point. This is the exact eutectic point where solid phase opx hasn't been exausted. In this phase region, temperature is exactly at the eutectic temperature,

$$\phi_f = \frac{\mathcal{H}}{L\rho_{f,1}} \quad \phi_2 = \frac{\mathcal{C} - \phi_f \rho_{f,1} \chi_e}{\rho_{2,1}} \quad \phi_1 = 1 - \phi_f - \phi_2 \quad \text{and} \quad \mathcal{T} = 0 \tag{6.6}$$

In this region, density of melt is fixed as eutectic density $\rho_f = \rho_e$.

6.1.4. Region IV. In this region, melting starts at the point left to the eutectic melting point. This is the melting happening in upper mantle. In this phase region, opx has exausted in solid matrix and temperature keeps rising. As more olivine melting, the density of melting changes.

$$\phi_2 = 0 \tag{6.7}$$

6.1.5. Region V. In this region, melting starts at the point right to the eutectic melting point. We are not investigate too much in this type of melting.

6.1.6. Region VI. In this region, only melting is present. Hence the volume fraction is given:

$$\phi_1 = \phi_2 = 0, \quad \phi_f = 1 \tag{6.8}$$

6.2. Further approximation and simplified phase split. To further simplify my calculation of phase diagram, I propose following approximations:

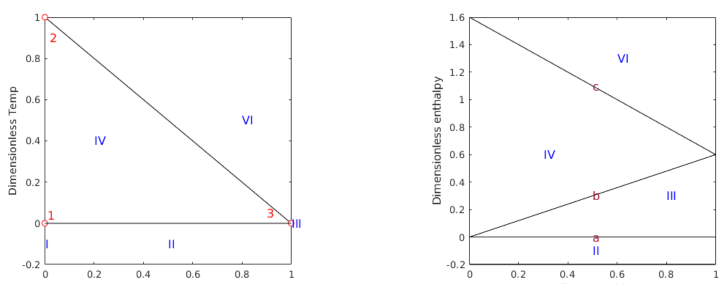
- We approximate both olivine and orthopyroxene with one uniform constant density.
- Heat capacity has a complex dependency on temperature, pressure and mixture. We assume the same specific heat capacity under circumstance of mantle melting.
- Ignore both thermal expansion and density change upon phase transition.
- Assume olivine and orthopyroxene share one same constant latent heat $L_1 = L_2 = L$.

Under such assumption, we can get a much simplified concentration and bulk enthalpy:

$$\begin{aligned} \mathcal{C} &= \phi_2 + \phi_f \chi_f(\mathcal{T}, \mathcal{P}) \\ \mathcal{H} &= (\phi_1 + \phi_2 + \phi_f)\mathcal{T} + \phi_f L_f \\ &= \mathcal{T} + \phi_f L \end{aligned} \tag{6.9}$$

6.2.1. Phase diagrams. In practical, we are focusing on eutectic melting happening when mass fraction of opx is left to eutectic point. We notice that upper mantle consists of more than 80% of olivine according to[8]. Hence, we only need to restrict our calculation to the left side of eutectic point.

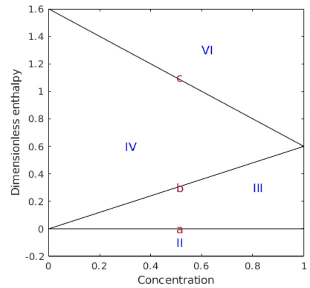
We can also plot a $\mathcal{H} - \chi$ diagram based on the $\mathcal{T} - \chi$ plot above.



More importantly, we need $\mathcal{H} - \mathcal{C}$ phase diagram. Now we calculate the line b separate region III (eutectic melting) and region IV (super-eutectic melting). Along line b, all opx has been exhausted and eutectic melting has stopped, hence $\phi_2 = 0$. The resulting $\mathcal{C} = \phi_f \chi_f$. According to our assumption of homogenous density, at the solidus region, we have the following relationship: $\chi = \frac{\phi_2}{\phi_2 + \phi_1} = \frac{M_2}{M_1 + M_2}$. Therefore, when all opx exhausted $\phi_2 = 0$, $\phi_f = X/X_e = \chi$. Along this line b, $\chi_f = \chi_e = 1$ is a constant for the fact that all the melting are eutectic melting. Hence along line b, we have

$$\mathcal{C} = \chi \tag{6.10}$$

Now let's turn to another line c separating region IV (super-eutectic melting) and region VI (super-liquidius region). On this line c, all solid has molten $\phi_f = 1$. The resulting $\mathcal{C} = \chi_f = \chi$. Hence the final $\mathcal{H} - \mathcal{C}$ diagram is:



6.2.2. Region II. Two-phase sub-solidus: olivine + opx. The conditions to be in region II are:

$$\mathcal{H} < 0 \tag{6.11}$$

In this phase region, temperature hasn't reached eutectic melting temperature.

$$\phi_2 = \mathcal{C} \quad \phi_1 = 1 - \phi_2 \quad \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \frac{\mathcal{H}}{\phi_1 + \phi_2} = \mathcal{H} < 0 \tag{6.12}$$

6.2.3. Region III. In this three-phase eutectic region, all three phases are present. In this phase region, temperature is exactly at the eutectic temperature,

$$\phi_f = \frac{\mathcal{H}}{L} \quad \phi_2 = \mathcal{C} - \phi_f \chi_f(\mathcal{T}, \mathcal{P}) \quad \phi_1 = 1 - \phi_f - \phi_2 \quad \text{and} \quad \mathcal{T} = 0 \tag{6.13}$$

Where $\chi_f(\mathcal{T}, \mathcal{P}) = \chi_e(\mathcal{P})$, all melting is eutectic melting. Mass fraction is fixed as eutectic mass fraction.

6.2.4. Region IV. In this super-eutectic two-phase region, opx has been exhausted in eutectic melting. The volumetric fractions can be determined as:

$$\phi_2 = 0 \quad \phi_f = \frac{\mathcal{C}}{\chi_f(\mathcal{T}, \mathcal{P})} \tag{6.14}$$

The corresponding temperature should be determined by solving the following equation.

$$F(\mathcal{T}) = \mathcal{H} - \mathcal{T} - \frac{\mathcal{C}}{\chi_f(\mathcal{T}, \mathcal{P})} L = 0 \tag{6.15}$$

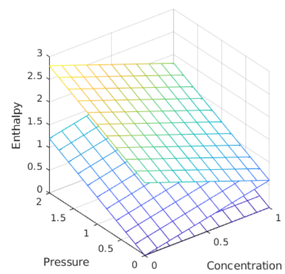
Simple Newton method can be applied to solve this equation if we apply nonlinear $\chi_f - \mathcal{T}$ relationship. If we use simple linear relationship between mass fraction and temperature as $\chi_f = \chi_e(1 - \mathcal{T})$, we can obtain a simple quadratic equation:

$$\begin{gathered} (1 - \mathcal{T})\mathcal{H} - (1 - \mathcal{T})\mathcal{T} - \mathcal{C}L/\chi_e(\mathcal{P}) = 0 \\ \mathcal{T}^2 - (\mathcal{H} + 1)\mathcal{T} + \mathcal{H} - \mathcal{C}L/\chi_e(\mathcal{P}) = 0 \end{gathered} \tag{6.16}$$

6.2.5. Region VI. In this single phase super-liquidus region, we can get volume fractions and temperature easily as:

$$\phi_1 = \phi_2 = 0 \quad \phi_f = 1 \quad \text{and} \quad \mathcal{T} = \mathcal{H} - L \tag{6.17}$$

6.3. Pressure and melting point. Now I take pressure change into account. In general, the higher the pressure, the higher the eutectic melting point. We assume a linear relationship involving nondimensionlized pressure.



7. Magma production. For our very first simulation, we would like to simulate the production of magma in a simplified controlled scenerio. I was inspired by the setup described previously in[?]. The simulation will be setup in a chunk of rock ascending with constant upwelling velocity. A linear adiabatic temperature profile is applied here. A linear intial rock composition is also applied here.

7.1. Mathematical model. We are going to show our coupled darcy-stokes semi-degenerate model can solve this problem well. In[?], matrix creeping (Stokes equation) did not directly involved in the simulation due to the fact that boundary condition restricts a constant background upwelling ascending velocity. The solid matrix pressure is therefore strictly lithostatic pressure relating only to depth. In[4], only energy transports while concentration does not participate in transport phenomenon.

$$\begin{aligned} \check{\mathbf{v}}_r + \phi_f^{1+\Theta}\nabla(\phi_f^{-1/2}\check{q}_f) &= 0 \\ \phi_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0 \\ \nabla\check{q} - \nabla\cdot\hat{\sigma}(\check{\mathbf{v}}_s) &= -(1-\phi_f)\hat{\mathbf{g}} \\ \nabla\cdot\check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f}(\check{q}_f - \phi_f^{1/2}\check{q}) &= 0 \end{aligned} \tag{7.1}$$

For simplification, we won't include diffusion into our transport model. We drop the diffusion term here for the fact that diffusion coefficient is relatively small compared to convective phenomenon. On the other hand, eutectic melting has a constant concentration such that gradient of concentration is of zero value. Hence the transport of component 2 (opx) can be expressed in terms of mixture theory as:

$$\begin{aligned} \bar{C}_t + \nabla\cdot(\overline{C\mathbf{v}}) &= 0 \\ \bar{C}_t + \nabla\cdot(\mathbf{v}_e^*\bar{C}) &= 0 \end{aligned} \tag{7.2}$$

where $\mathbf{v}_e^*$ denotes effective veclocity $\mathbf{v}_e^* = \kappa\mathbf{v}_f + (1-\kappa)\mathbf{v}_s$ and $\kappa = \frac{\phi_f C_f}{\bar{C}}$ where C_f and $\bar{C}$ using values from previous time step. Enthalpy is transported in a similar manner.

$$\bar{\mathcal{H}}_t + \nabla\cdot(\overline{\mathcal{H}\mathbf{v}} - \bar{K}\nabla\mathcal{T}) = 0 \tag{7.3}$$

7.1.1. Boundary conditions.

- Prescribe essential boundary condition of constant upwelling ascending velocity all around the computational domain. $\mathbf{v}_s = \mathbf{W}$
-

7.1.2. Initial conditions.

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